CSV 溫度變化分析

taiwan_temps_yearly_s...
Spreadsheet

I'd like to summarize this CSV. Please give me the average change in temperature for each station, for the last 30 years. The catch is that the reporting years are different for each station. Please ignore any years before 1985.

I interpreted your request as follows:

- Ignore all records before **1985**.
- For each station, use the **last 30 reporting years available for that station** (or all available years after 1985 if there are fewer than 30).
- Compute the **average year-to-year change** in `avg_of_monthly_max_c` over that period (i.e., the mean of consecutive annual differences).

Here are the results:

Station	Reporting period used	Years	Average annual change (°C/year)
新北 (Xinbei)	2023–2026	4	-0.3179
淡水 (Danshui)	1997–2026	30	-0.0014
鞍部 (Anbu)	1997–2026	30	-0.0097
臺北 (Taipei)	1997–2026	30	0.0421
竹子湖 (Zhuzihu)	1997–2026	30	-0.0447
基隆 (Keelung)	1997–2026	30	0.0343
彭佳嶼 (Pengjiayu)	1997–2026	30	-0.0037
花蓮 (Hualien)	1997–2026	30	0.0109
新屋 (Xinwu)	2013–2026	14	-0.1841
宜蘭 (Yilan)	1997–2026	30	-0.0015
金門 (Kinmen)	2004–2026	23	-0.0045

Station	Reporting period used	Years	Average annual change (°C/year)
田中 (Tianzhong)	2020–2026	7	-0.1461
後龍 (Houlong)	2024–2026	3	-0.9643
古坑 (Gukeng)	2024–2026	3	1.5000
東吉島 (Dongji Island)	1997–2026	30	0.0433
澎湖 (Penghu)	1997–2026	30	0.0007
臺南 (Tainan)	1997–2026	27	-0.0136
永康 (Yongkang)	1997–2026	30	0.0195
高雄 (Kaohsiung)	2022–2026	5	-0.0977
嘉義 (Chiayi)	1997–2026	30	0.0222
臺中 (Taichung)	1997–2026	30	0.0372
阿里山 (Alishan)	1997–2026	30	-0.0333
大武 (Dawu)	1997–2026	30	0.0583
玉山 (Yushan)	1997–2026	30	0.1234
新竹 (Hsinchu)	1997–2026	30	0.0434
恆春 (Hengchun)	1997–2026	30	0.0025
成功 (Chenggong)	1997–2026	30	0.0145
蘭嶼 (Lanyu)	1997–2026	30	-0.0224
日月潭 (Sun Moon Lake)	1997–2026	30	0.0278
臺東 (Taitung)	1997–2026	30	-0.0112

A note on interpretation: I calculated the **average of the year-to-year temperature changes**. If instead you meant the **overall trend** (for example, the slope from a linear

regression over the last 30 years), ↪ I can compute that instead, which is often a more robust measure of long-term warming.

1/2

Compute the overall trend using the slope from a linear regression over the last 30 years

Using your requested approach, I:

- Ignored all years **before 1985**.
- For each station, selected the **last 30 reporting years available** (or all available years if fewer than 30).
- Fit a simple linear regression of annual average maximum temperature versus year:
  - **Temperature = intercept + slope × year**
- Reported the **slope** in **°C per year**.

Station	Period used	Years	Trend (°C/year)
古坑 (Gukeng)	2024–2026	3	**1.5000**
嘉義 (Chiayi)	1997–2026	30	**0.0424**
基隆 (Keelung)	1997–2026	30	**0.0075**
大武 (Dawu)	1997–2026	30	**0.0157**
宜蘭 (Yilan)	1997–2026	30	**0.0114**
彭佳嶼 (Pengjiayu)	1997–2026	30	**−0.0007**
後龍 (Houlong)	2024–2026	3	**−0.9643**
恆春 (Hengchun)	1997–2026	30	**−0.0033**
成功 (Chenggong)	1997–2026	30	**−0.0128**
新北 (Xinbei)	2023–2026	4	**−0.3177**
新屋 (Xinwu)	2013–2026	14	**−0.0780**
新竹 (Hsinchu)	1997–2026	30	**0.0256**
日月潭 (Sun Moon Lake)	1997–2026	30	**0.0350**

Station	Period used	Years	Trend (°C/year)
東吉島 (Dongji Island)	1997–2026	30	**0.0447**
永康 (Yongkang)	1997–2026	30	**0.0091**
淡水 (Danshui)	1997–2026	30	**0.0235**
澎湖 (Penghu)	1997–2026	30	**−0.0078**
玉山 (Yushan)	1997–2026	30	**0.1462**
田中 (Tianzhong)	2020–2026	7	**−0.0561**
竹子湖 (Zhuzihu)	1997–2026	30	**−0.0417**
臺中 (Taichung)	1997–2026	30	**0.0258**
臺北 (Taipei)	1997–2026	30	**0.0272**
臺南 (Tainan)	1994–2026	30	**−0.0070**
臺東 (Taitung)	1997–2026	30	**−0.0269**
花蓮 (Hualien)	1997–2026	30	**−0.0065**
蘭嶼 (Lanyu)	1997–2026	30	**−0.0059**
金門 (Kinmen)	2004–2026	23	**0.0792**
阿里山 (Alishan)	1997–2026	30	**−0.0088**
鞍部 (Anbu)	1997–2026	30	**−0.0287**
高雄 (Kaohsiung)	2022–2026	5	**−0.1848**

A few notes:

- Stations with **only 3–7 years** of data (e.g. Gukeng, Houlong, Xinbei, Kaohsiung, Tianzhong) have highly uncertain slopes because they are based on short time series.
- Among stations with a full 30-year record, **Yushan** shows the strongest positive trend (**+0.1462 °C/year**), while **Zhuzihu** (**−0.0417 °C/year**) and **Anbu** (**−0.0287 °C/year**) have the strongest negative trends.